HEALTHCARE DATA ANALYSIS AND DISEASE RISK PREDICTION

Exploratory Data Analysis, Visualization and Data Preprocessing

Project Report

Prepared using the supplied healthcare datasets, Google Colab notebook and visualization outputs.

Project domain: Healthcare Analytics / Artificial Intelligence and Data Science

Abstract

This project, **Healthcare Data Analysis and Disease Risk Prediction**, examines patient health indicators to explore patterns that may be associated with disease categories. The supplied data contains 15 records and eight columns: Age, Gender, BMI, Smoking, Glucose, BP, Cholesterol and Disease. The workflow compares a raw healthcare spreadsheet with a cleaned spreadsheet, then uses exploratory data analysis (EDA) and visualizations to examine patient age distribution, average BMI and glucose by disease category, blood pressure against cholesterol, disease distribution by gender, and correlations among features.

The charts help summarize relationships in the sample data, but the dataset is small and should be treated as an educational demonstration. Correlation does not establish causation, and the visualizations do not provide a clinically validated diagnosis or prediction. A reliable predictive system would require a larger, representative dataset, a defined target label, model training and evaluation, and appropriate clinical review.

1. Introduction

Healthcare datasets can contain useful information about patient characteristics and health measurements, but raw records may include missing values, inconsistent labels and mixed data types. Data analysis makes these records easier to inspect and can reveal patterns that merit further investigation. This project uses Python-based data handling and visualization to demonstrate a basic healthcare analytics workflow.

2. Problem Statement

Patient records can be difficult to interpret when data is incomplete or inconsistent. The project addresses this by preparing the healthcare data, summarizing key indicators, and visualizing how those indicators vary across the available disease categories and demographic groups.

3. Objectives

• Inspect raw and cleaned healthcare spreadsheets and identify missing values.

• Use Pandas to examine the dataset structure, data types and descriptive statistics.

• Visualize the distribution of patient age and compare BMI and glucose across disease categories.

• Explore blood pressure and cholesterol values by smoking status and gender.

• Use a correlation heatmap to review pairwise linear associations among numeric variables.

• Document limitations and outline what is needed before developing a dependable disease-risk prediction model.

4. Dataset Description

The supplied Excel files are *messy_healthcare_data.xlsx* and *cleaned_healthcare_data.xlsx*. Each contains 15 rows and eight columns. The raw file uses readable categories such as Male/Female, Yes/No and disease names, while the cleaned file uses numeric encodings for Gender, Smoking and Disease.

Column	Meaning in this project	Representation in cleaned file
Age	Patient age	Numeric
Gender	Gender category	0 / 1
BMI	Body mass index	Numeric
Smoking	Smoking-status category	0 / 1
Glucose	Glucose measurement	Numeric

Column	Meaning in this project	Representation in cleaned file
BP	Blood pressure value	Numeric
Cholesterol	Cholesterol measurement	Numeric
Disease	Disease category/label	Numeric category

The numeric category codes are labels, not necessarily quantities. The exact meaning of each encoded value should be documented in a data dictionary before modeling. The supplied project material does not establish that the disease categories are ordered severity stages.

5. Tools and Technologies

• Python in Google Colab for running the analysis notebook.

• Pandas for reading Excel files, inspecting records and handling tabular data.

• Matplotlib and Seaborn for charts and statistical visualization.

• Excel spreadsheets as the raw and cleaned data sources.

6. Methodology

6.1 Load and inspect

The notebook reads the Excel files using pandas.read_excel(), then checks the dataset shape, column information and descriptive statistics. The raw and cleaned spreadsheets are compared using shape and missing-value counts.

6.2 Data preprocessing

The project distinguishes the raw file from the cleaned file and encodes categorical variables numerically for analysis. A robust preprocessing stage should also check missing values, duplicate rows, valid measurement ranges and consistent category mappings. The report does not assume that every possible validation step was performed unless it is shown in the supplied notebook or files.

6.3 Exploratory data analysis

Five supplied charts are used to inspect the sample: age distribution, mean BMI and glucose by disease category, BP versus cholesterol grouped by smoking and gender, disease counts across gender, and a correlation heatmap.

6.4 Prediction scope

The project title includes disease-risk prediction, and the presentation describes prediction as a future feature. The supplied charts and spreadsheet alone do not demonstrate a trained, evaluated machine-learning model. To claim predictive performance, a model would need to be trained on suitable labelled data and evaluated using held-out data and appropriate metrics.

7. Results and Visualization

Age distribution of patients

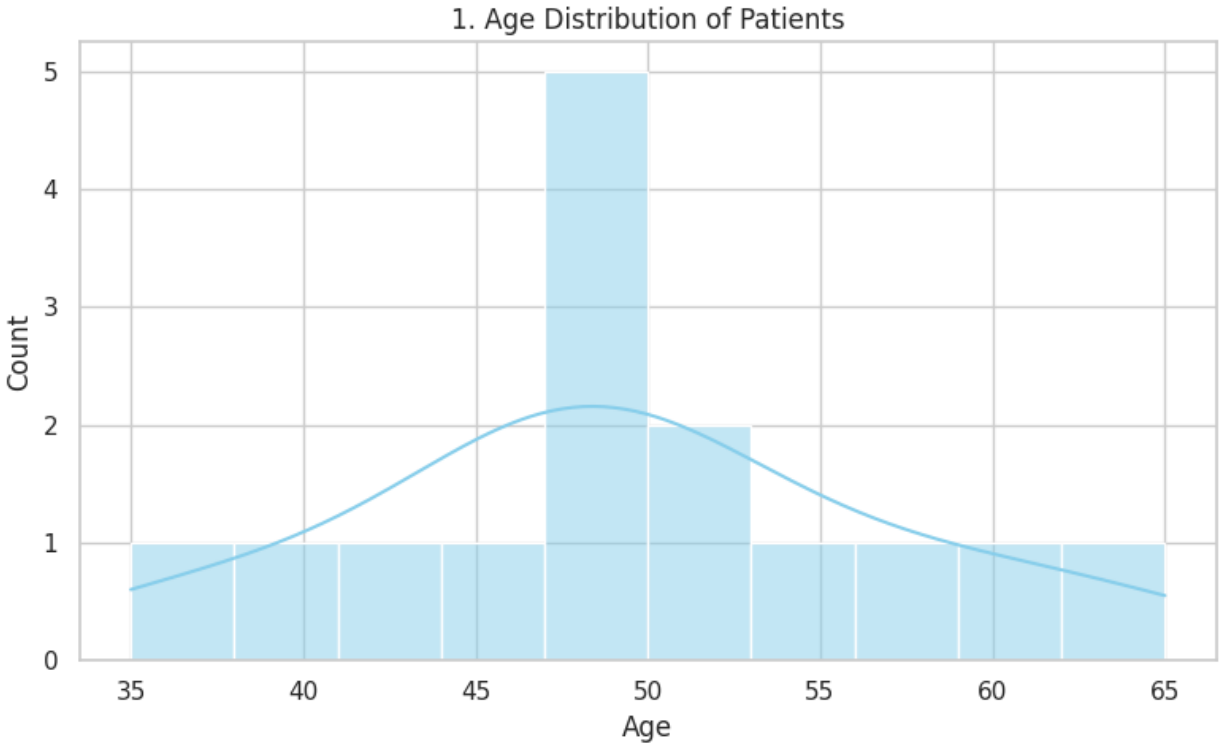


Figure 1. Age distribution of patients

The histogram shows the spread of patient ages in this sample. The largest bin is around the late 40s, with observations extending approximately from the mid-30s to the mid-60s.

Average BMI and glucose by disease category

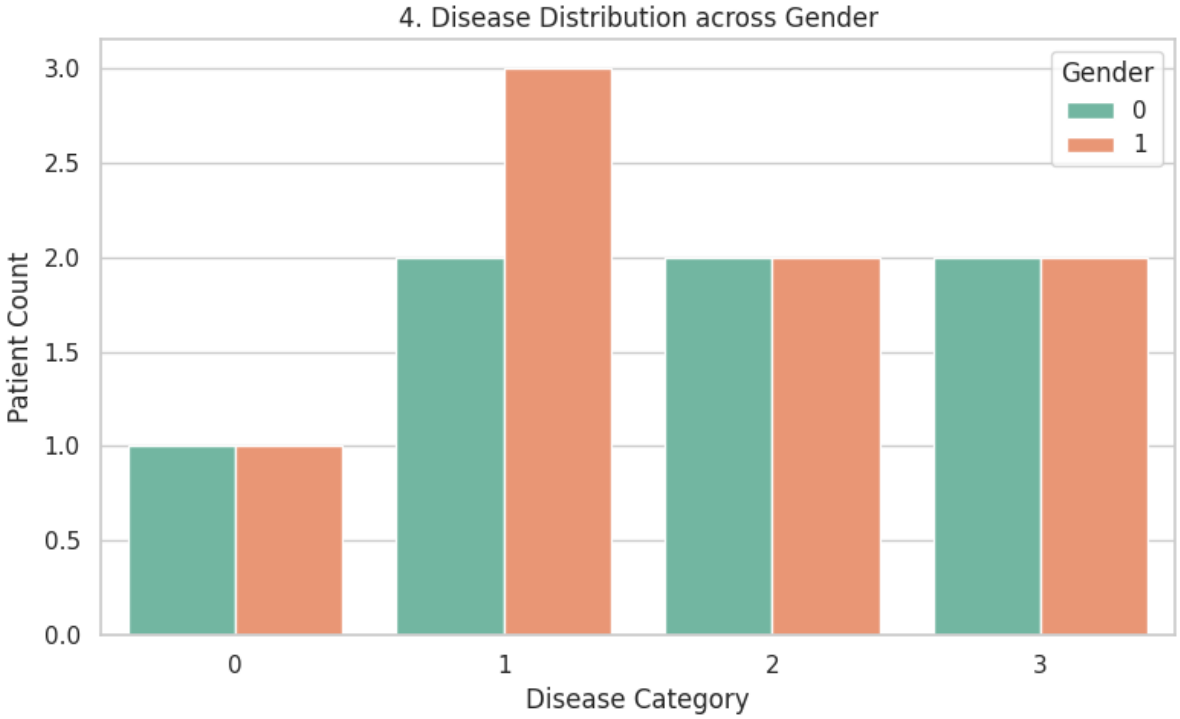


Figure 2. Average BMI and glucose by disease category

The grouped bars compare the average BMI and glucose values for each encoded disease category. In this sample, the glucose bar is highest for category 2; the interpretation depends on the category mapping and the small number of records.

Blood pressure versus cholesterol by smoking status and gender

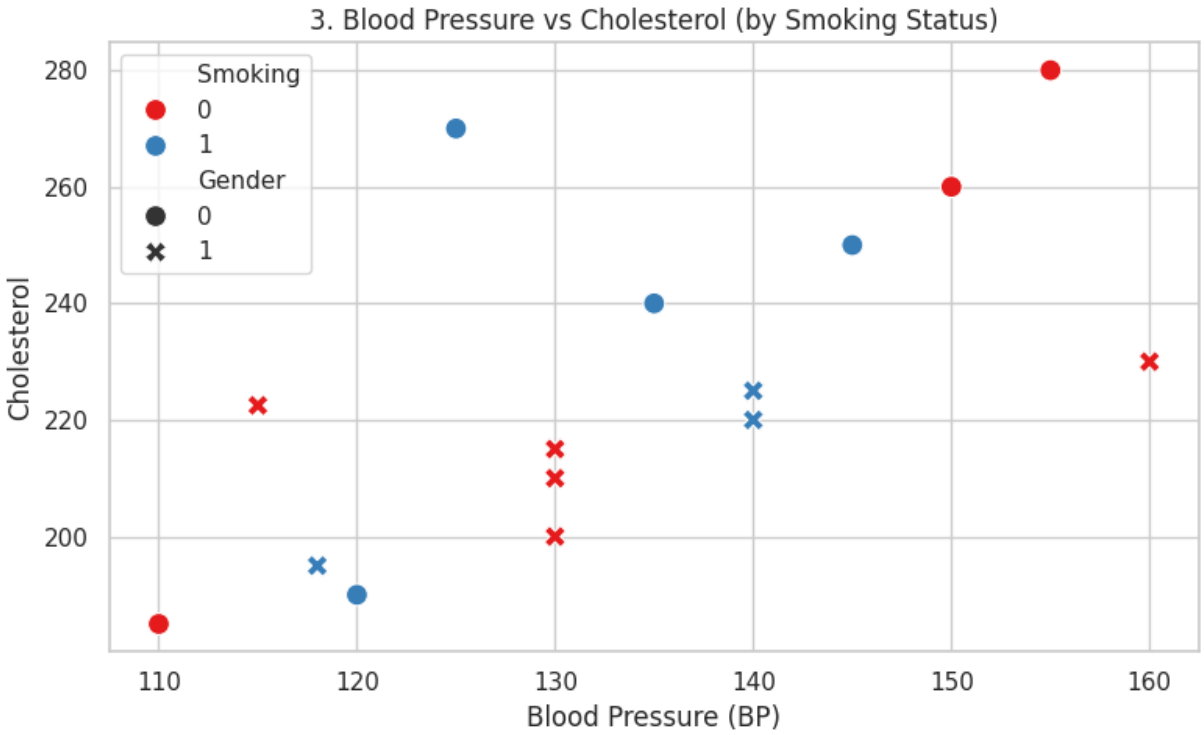


Figure 3. Blood pressure versus cholesterol by smoking status and gender

Each point represents a record. The chart helps compare BP and cholesterol values across the encoded smoking and gender groups. The small sample is not sufficient to infer a causal relationship.

Disease distribution across gender

<Figure size 800x500 with 0 Axes>

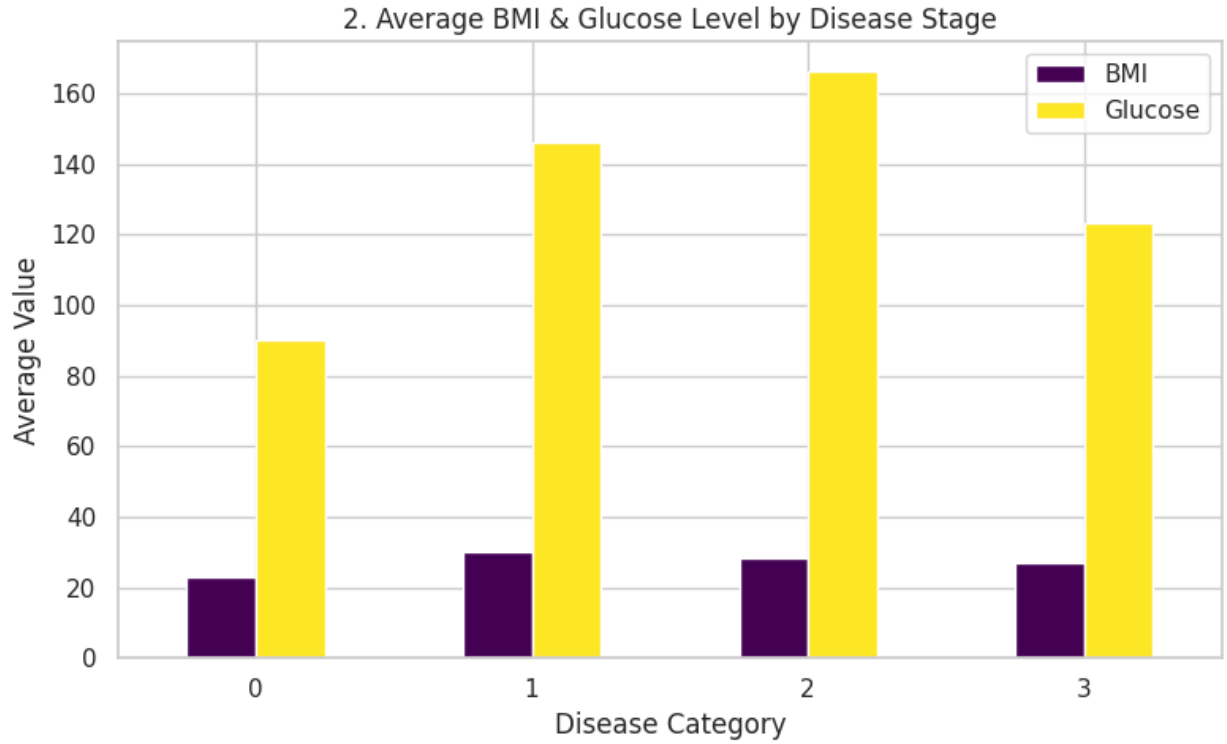


Figure 4. Disease distribution across gender

The count plot compares record counts by disease category and gender code. Some categories have similar counts, while category 1 has more records for gender code 1 in the displayed sample.

Correlation heatmap of healthcare features

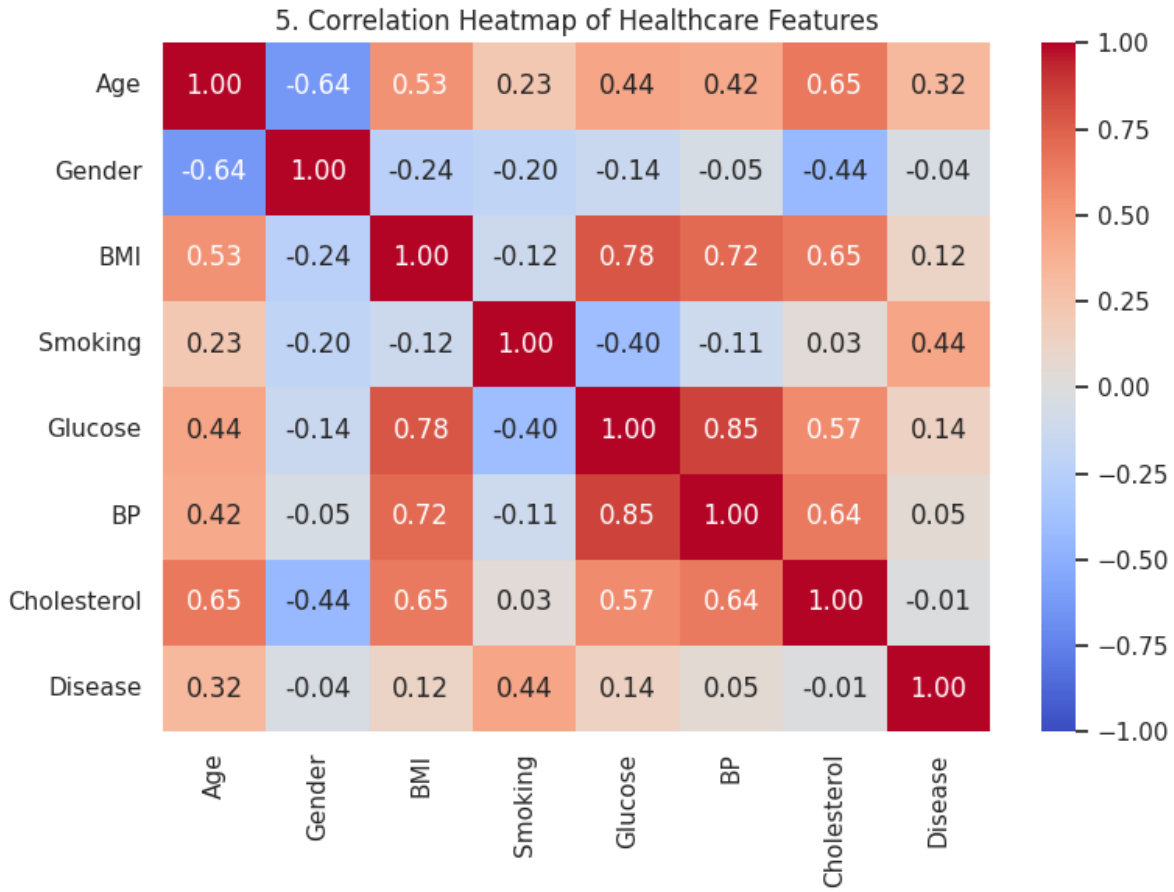


Figure 5. Correlation heatmap of healthcare features

The heatmap displays pairwise correlation coefficients. Glucose and BP have a strong positive correlation in the shown matrix (0.85); BMI and glucose are also positively correlated (0.78). These are associations in the sample, not proof of causation or medical risk.

8. Key Findings

• The sample contains 15 patient records with eight variables.

• Patient age values in the plotted sample range approximately from 35 to 65 years.

• Average glucose varies noticeably across the encoded disease categories in the grouped bar chart.

• The correlation matrix shows positive associations between glucose and BP (0.85), BMI and glucose (0.78), and BMI and BP (0.72).

• The disease-by-gender chart displays uneven counts in some categories; counts should be interpreted in light of the small sample size.

• The plots are useful for exploratory understanding, but they cannot establish causality or provide clinical conclusions.

9. Limitations

• Only 15 records are present, so the sample is too small for reliable generalization.

• The meanings of numeric Gender, Smoking and Disease codes should be documented explicitly.

• Correlation coefficients may be unstable in a small dataset and can be influenced by individual records.

• The visualizations do not validate a machine-learning model or its predictive performance.

• Health measurements and disease labels must be checked against their source definitions and units before real-world use.

10. Future Enhancements

• Collect a larger, representative and ethically sourced dataset with clear definitions and units.

• Create a documented preprocessing pipeline for missing values, duplicates, encoding and range checks.

• Define the prediction target clearly and prevent data leakage between training and test sets.

• Train and compare suitable classification models, using stratified validation where appropriate.

• Evaluate with metrics such as precision, recall, F1-score, ROC-AUC and a confusion matrix, chosen to match the task and class balance.

• Add explainability, privacy safeguards and review by qualified healthcare professionals before any real-world deployment.

11. Conclusion

This project demonstrates a basic workflow for healthcare data analysis: inspect raw and cleaned records, summarize patient indicators, and use plots to explore distributions and relationships. The supplied figures show variation in age, glucose, BMI, blood pressure, cholesterol and disease-category counts. The strongest displayed correlations include glucose with BP and BMI with glucose. These results are exploratory only. A dependable disease-risk prediction system requires more data, documented labels, a properly trained and evaluated model, and medical oversight.

References / Project Materials

• User-supplied Google Colab notebook: Untitled4.ipynb.

• User-supplied spreadsheet: messy_healthcare_data.xlsx.

• User-supplied spreadsheet: cleaned_healthcare_data.xlsx.

• User-supplied visualization images: age distribution, grouped bar charts, scatter plot and correlation heatmap.

• User-supplied presentation: Healthcare_Data_Analysis_Disease_Risk_Prediction.pptx.

Note: This is an academic analytics report, not medical advice or a diagnostic document.